

CUSTOMER CHURN PREDICTION SYSTEM

Project Documentation Report

Submitted By:

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CERTIFICATE

This is to certify that the project titled "**Customer Churn Prediction System**" has been successfully completed by **Shubham Mishra** as part of the Data Science and Machine Learning learning program.

This project demonstrates the practical implementation of Machine Learning algorithms, data preprocessing, exploratory data analysis, model evaluation, explainable AI, and deployment using Streamlit.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me during the development of this project. I am thankful to my mentors and instructors for providing valuable guidance and knowledge related to Data Science and Machine Learning.

I would also like to thank my friends and family for their continuous support and encouragement throughout the project development process.

Finally, I would like to thank the open-source community and online resources that helped me complete this project successfully.

ABSTRACT

Customer churn prediction is one of the most important business applications of Machine Learning in the telecom industry. Customer churn refers to the situation when customers stop using a company's services. Retaining existing customers is much more cost-effective than acquiring new customers.

The main objective of this project is to build a Machine Learning model capable of predicting whether a customer is likely to churn based on customer demographic details, account information, and service usage patterns.

This project includes:

- Data preprocessing and cleaning
- Exploratory Data Analysis (EDA)
- Feature engineering
- Machine Learning model training
- Model evaluation
- SHAP explainability analysis
- Streamlit web application deployment

Different Machine Learning algorithms such as Logistic Regression, Random Forest, and XGBoost were used for classification. The project also provides business insights that can help telecom companies improve customer retention strategies.

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1. INTRODUCTION

Machine Learning has become an important technology for solving real-world business problems. One of the major challenges faced by telecom companies is customer churn. Customer churn directly impacts company revenue and business growth.

Customer churn prediction helps organizations identify customers who are likely to leave the company. By identifying such customers early, businesses can take preventive actions such as special offers, discounts, or improved customer service.

In this project, a Machine Learning-based Customer Churn Prediction System has been developed using Python and various Data Science libraries.

The project uses customer-related data such as:

- Gender
- Tenure
- Internet service
- Monthly charges
- Contract type
- Payment methods

These features are used to train Machine Learning models capable of predicting customer churn.

2. PROBLEM STATEMENT

Telecom companies lose a significant number of customers every year due to churn. It is difficult to manually identify customers who are likely to leave.

The problem statement of this project is:

“To develop a Machine Learning model that can accurately predict whether a customer will churn based on customer service and account information.”

3. OBJECTIVES

The main objectives of this project are:

- To analyze customer behavior using data analytics.
 - To preprocess and clean customer data.
 - To perform exploratory data analysis.
 - To build multiple Machine Learning models.
 - To compare model performance.
 - To explain predictions using SHAP.
 - To create a user-friendly Streamlit web application.
 - To generate business insights for customer retention.
-

4. DATASET DESCRIPTION

The dataset used in this project is the **Telco Customer Churn Dataset**.

Dataset Source: Kaggle

The dataset contains information related to telecom customers.

Features in Dataset

Feature	Description
gender	Customer gender
SeniorCitizen	Whether customer is senior citizen
Partner	Whether customer has partner
Dependents	Whether customer has dependents
tenure	Number of months customer stayed
PhoneService	Phone service status
InternetService	Type of internet service
Contract	Contract type
MonthlyCharges	Monthly bill amount
TotalCharges	Total bill amount
Churn	Target variable

Total records in dataset: 7043

5. TECHNOLOGIES USED

Programming Language

- Python

Libraries Used

Library	Purpose
Pandas	Data handling

Library	Purpose
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Scikit-learn	Machine Learning models
XGBoost	Advanced boosting model
SHAP	Explainable AI
Streamlit	Web application
Pickle	Model saving

6. SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB minimum
- Storage: 1 GB free space

Software Requirements

- Python 3.x
 - VS Code / Jupyter Notebook
 - Streamlit
 - Git and GitHub
-

7. METHODOLOGY

The project follows the following workflow:

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis
4. Feature Encoding
5. Train-Test Split
6. Model Training
7. Model Evaluation
8. SHAP Explainability

- 9. Model Saving
 - 10. Streamlit Application Development
 - 11. GitHub Deployment
-

8. DATA PREPROCESSING

Data preprocessing is one of the most important stages of any Machine Learning project.

The following preprocessing steps were performed:

8.1 Removing Unnecessary Columns

The customerID column was removed because it does not contribute to prediction.

8.2 Handling Missing Values

The TotalCharges column contained missing values. These values were converted into numeric format and missing values were filled using the median.

8.3 Encoding Categorical Variables

Machine Learning algorithms cannot understand text values. Therefore, Label Encoding was applied to categorical columns.

Examples:

- Male → 1
 - Female → 0
 - Yes → 1
 - No → 0
-

9. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis helps understand patterns and relationships within data.

9.1 Churn Distribution

A count plot was used to analyze customer churn distribution.

Observation: Most customers stayed with the company, while a smaller percentage churned.

9.2 Contract Type vs Churn

Customers with month-to-month contracts were more likely to churn.

9.3 Monthly Charges Analysis

Customers with higher monthly charges showed higher churn probability.

9.4 Tenure Analysis

Customers with lower tenure were more likely to leave.

9.5 Correlation Heatmap

A heatmap was created to analyze correlations between numerical features.

10. FEATURE ENGINEERING

Feature engineering was performed to improve model performance.

Categorical variables were transformed into numerical values using Label Encoding.

Features and target variables were separated:

- $X \rightarrow$ Independent features
- $y \rightarrow$ Target variable (Churn)

The dataset was split into:

- 80% Training Data
 - 20% Testing Data
-

11. MACHINE LEARNING MODELS

Three Machine Learning models were trained and evaluated.

11.1 Logistic Regression

Logistic Regression is a basic classification algorithm used for binary classification problems.

Advantages:

- Simple
- Fast
- Easy to interpret

Accuracy Achieved: 82%

11.2 Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees.

Advantages:

- Better accuracy
- Handles overfitting well
- Robust performance

Accuracy Achieved: 80%

11.3 XGBoost Classifier

XGBoost is an advanced boosting algorithm widely used in Machine Learning competitions.

Advantages:

- High performance
- Efficient computation
- Better feature importance analysis

Accuracy Achieved: 81%

12. MODEL EVALUATION

Different evaluation metrics were used to evaluate model performance.

12.1 Accuracy Score

Measures overall correctness of predictions.

12.2 Classification Report

Includes:

- Precision
- Recall
- F1-score

12.3 Confusion Matrix

Used to visualize correct and incorrect predictions.

12.4 ROC-AUC Score

Measures classification performance.

13. SHAP EXPLAINABILITY

SHAP (SHapley Additive exPlanations) was used to explain Machine Learning predictions.

SHAP helps identify:

- Which features influence churn most
- Positive and negative feature impacts
- Model interpretability

Important features identified:

- MonthlyCharges
- tenure
- Contract
- InternetService

SHAP visualizations included:

- Beeswarm plot
 - Feature importance plot
-

14. STREAMLIT WEB APPLICATION

A Streamlit web application was developed for real-time churn prediction.

Features of Application

- User-friendly interface
- Interactive input fields
- Real-time prediction
- Instant churn result display

Users can enter customer information and predict whether the customer is likely to churn.

15. BUSINESS INSIGHTS

The project generated important business insights.

Key Insights

- Customers with high monthly charges are more likely to churn.
- Customers with short tenure tend to leave quickly.
- Month-to-month contracts have higher churn rates.
- Long-term contracts reduce churn probability.

Business Benefits

- Improved customer retention
 - Reduced revenue loss
 - Better customer targeting
 - Improved decision-making
-

16. RESULTS

Final Model Performance

Model	Accuracy
Logistic Regression	82%
Random Forest	80%
XGBoost	81%

The Logistic Regression model achieved the highest accuracy in this project.

17. ADVANTAGES

- Predicts customer churn effectively
 - Helps telecom companies improve retention
 - Provides explainable AI insights
 - User-friendly web interface
 - Real-world business application
-

18. LIMITATIONS

- Model accuracy can be improved further
 - Limited dataset size
 - Only telecom dataset used
 - Hyperparameter tuning was limited
-

19. FUTURE SCOPE

Future improvements may include:

- Hyperparameter tuning
 - Deep Learning models
 - Real-time cloud deployment
 - Larger datasets
 - Advanced feature engineering
 - Integration with live telecom systems
-

20. CONCLUSION

The Customer Churn Prediction System successfully demonstrates the practical implementation of Machine Learning for solving business problems.

The project involved complete Data Science workflow including:

- Data preprocessing
- Data visualization
- Machine Learning model development
- Model evaluation
- Explainable AI
- Web application deployment

The system can help telecom companies identify customers likely to churn and take preventive actions to improve customer retention.

Overall, this project provided valuable hands-on experience in Data Science, Machine Learning, and deployment technologies.

21. REFERENCES

1. Kaggle Telco Customer Churn Dataset
 2. Scikit-learn Documentation
 3. XGBoost Documentation
 4. SHAP Documentation
 5. Streamlit Documentation
 6. Python Official Documentation
-

APPENDIX

Screenshots Included

- Main Application Interface
 - Prediction Result
 - Accuracy Comparison Graph
 - Confusion Matrices
 - SHAP Explainability Graphs
-

THANK YOU

Jupyter Notebook PDF Export

Code Cell 1

```
print("Jupyter Notebook Working Successfully")
```

Output:

```
Jupyter Notebook Working Successfully
```

Code Cell 2

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import (
    accuracy_score,
    classification_report,
    confusion_matrix,
    roc_auc_score
)

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

import pickle
import shap
```

Code Cell 3

```
df = pd.read_csv("../data/WA_Fn-UseC_-Telco-Customer-Churn.csv")
```

Code Cell 4

```
df.head()
```

Code Cell 5

```
df.info()
```

Output:

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype  
---  --
 0   customerID            7043 non-null  str    
 1   gender                7043 non-null  str    
 2   SeniorCitizen         7043 non-null  int64  
 3   Partner               7043 non-null  str    
 4   Dependents            7043 non-null  str
```

```
5 tenure 7043 non-null int64
6 PhoneService 7043 non-null str
7 MultipleLines 7043 non-null str
8 InternetService 7043 non-null str
9 OnlineSecurity 7043 non-null str
10 OnlineBackup 7043 non-null str
11 DeviceProtection 7043 non-null str
12 TechSupport 7043 non-null str
13 StreamingTV 7043 non-null str
14 StreamingMovies 7043 non-null str
15 Contract 7043 non-null str
16 PaperlessBilling 7043 non-null str
17 PaymentMethod 7043 non-null str
18 MonthlyCharges 7043 non-null float64
19 TotalCharges 7043 non-null str
20 Churn 7043 non-null str
dtypes: float64(1), int64(2), str(18)
memory usage: 1.9 MB
```

Code Cell 6

```
df.isnull().sum()
```

Code Cell 7

```
df.drop("customerID", axis=1, inplace=True)
```

Code Cell 8

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
```

Code Cell 9

```
df["TotalCharges"] = df["TotalCharges"].fillna(df["TotalCharges"].median())
```

Code Cell 10

```
df.isnull().sum()
```

Code Cell 11

```
sns.countplot(x="Churn", data=df)
plt.title("Customer Churn Distribution")
plt.show()
```

Markdown Cell 12

Insight:

Most customers stayed with the company, but a significant number of customers churned. This indicates the importance of customer retention strategies.

Code Cell 13

```
sns.countplot(x="Contract", hue="Churn", data=df)
plt.title("Contract Type vs Churn")
plt.xticks(rotation=45)
plt.show()
```

Markdown Cell 14

Insight:

Customers with month-to-month contracts are more likely to leave the company compared to customers with yearly contracts.

Code Cell 15

```
sns.histplot(df["MonthlyCharges"], kde=True)
plt.title("Monthly Charges Distribution")
plt.show()
```

Code Cell 16

```
sns.boxplot(x="Churn", y="MonthlyCharges", data=df)
plt.title("Monthly Charges vs Churn")
plt.show()
```

Code Cell 17

```
sns.boxplot(x="Churn", y="tenure", data=df)
plt.title("Tenure vs Churn")
plt.show()
```

Code Cell 18

```
plt.figure(figsize=(15,10))
sns.heatmap(df.corr(numeric_only=True),
            cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

Markdown Cell 19

EDA Summary

- Customers with high monthly charges tend to churn more.
- Customers with shorter tenure are more likely to leave.
- Month-to-month contracts have higher churn rates.
- Long-term contracts improve customer retention.

Code Cell 20

```
from sklearn.preprocessing import LabelEncoder

# Create encoder
le = LabelEncoder()

# Encode all object columns
for col in df.select_dtypes(include='object').columns:

    df[col] = le.fit_transform(df[col])
```

Output:

```
C:\Users\SHUBHAM MISHRA\AppData\Local\Temp\ipykernel_17568\517900884.py:7: Pandas4Warning: For backward compatibility
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details
    for col in df.select_dtypes(include='object').columns:
```

Code Cell 21

```
df.head()
```

Code Cell 22

```
X = df.drop("Churn", axis=1)
y = df["Churn"]
```

Code Cell 23

```
print(X.shape)
print(y.shape)
```

Output:

```
(7043, 19)
(7043,)
```

Code Cell 24

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
```



```
y,  
test_size=0.2,  
random_state=42  
)
```

Code Cell 25

```
print(X_train.shape)  
print(X_test.shape)
```

Output:

```
(5634, 19)  
(1409, 19)
```

Code Cell 26

```
from sklearn.linear_model import LogisticRegression  
lr = LogisticRegression(max_iter=5000)  
lr.fit(X_train, y_train)
```

Code Cell 27

```
lr_pred = lr.predict(X_test)
```

Code Cell 28

```
from sklearn.metrics import accuracy_score  
print("Accuracy:", accuracy_score(y_test, lr_pred))
```

Output:

```
Accuracy: 0.8168914123491838
```

Code Cell 29

```
from sklearn.metrics import classification_report  
print(classification_report(y_test, lr_pred))
```

Output:

	precision	recall	f1-score	support
0	0.86	0.90	0.88	1036
1	0.68	0.58	0.63	373
accuracy			0.82	1409
macro avg	0.77	0.74	0.75	1409
weighted avg	0.81	0.82	0.81	1409

Code Cell 30

```
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, lr_pred)

sns.heatmap(cm,
            annot=True,
            fmt='d')

plt.title("Confusion Matrix")

plt.show()
```

Code Cell 31

```
from sklearn.metrics import roc_auc_score

score = roc_auc_score(y_test, lr_pred)

print("ROC-AUC Score:", score)
```

Output:

```
ROC-AUC Score: 0.7416569192708604
```

Code Cell 32

```
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

rf.fit(X_train, y_train)
```

Code Cell 33

```
rf_pred = rf.predict(X_test)
```

Code Cell 34

```
print("Random Forest Accuracy:")

print(accuracy_score(y_test, rf_pred))
```

Output:

```
Random Forest Accuracy:
0.7955997161107168
```

Code Cell 35

```
print(classification_report(y_test, rf_pred))
```

Output:

	precision	recall	f1-score	support
0	0.83	0.91	0.87	1036

	1	0.66	0.47	0.55	373
accuracy				0.80	1409
macro avg	0.74	0.69	0.71		1409
weighted avg	0.78	0.80	0.78		1409

Code Cell 36

```
cm_rf = confusion_matrix(y_test, rf_pred)

sns.heatmap(cm_rf,
            annot=True,
            fmt='d')

plt.title("Random Forest Confusion Matrix")

plt.show()
```

Code Cell 37

```
from xgboost import XGBClassifier

xgb = XGBClassifier(
    n_estimators=100,
    learning_rate=0.1,
    max_depth=5,
    random_state=42
)

xgb.fit(X_train, y_train)
```

Code Cell 38

```
xgb_pred = xgb.predict(X_test)
```

Code Cell 39

```
print("XGBoost Accuracy:")

print(accuracy_score(y_test, xgb_pred))
```

Output:

```
XGBoost Accuracy:
0.8062455642299503
```

Code Cell 40

```
print(classification_report(y_test, xgb_pred))
```

Output:

	precision	recall	f1-score	support
0	0.84	0.90	0.87	1036
1	0.67	0.54	0.60	373
accuracy			0.81	1409
macro avg	0.76	0.72	0.73	1409

weighted avg	0.80	0.81	0.80	1409
--------------	------	------	------	------

Code Cell 41

```
cm_xgb = confusion_matrix(y_test, xgb_pred)

sns.heatmap(cm_xgb,
            annot=True,
            fmt='d')

plt.title("XGBoost Confusion Matrix")

plt.show()
```

Code Cell 42

```
models = pd.DataFrame({
    "Model": [
        "Logistic Regression",
        "Random Forest",
        "XGBoost"
    ],
    "Accuracy": [
        accuracy_score(y_test, lr_pred),
        accuracy_score(y_test, rf_pred),
        accuracy_score(y_test, xgb_pred)
    ]
})

models
```

Code Cell 43

```
sns.barplot(
    x="Model",
    y="Accuracy",
    data=models
)

plt.title("Model Accuracy Comparison")

plt.show()
```

Code Cell 44

```
import shap
```

Code Cell 45

```
explainer = shap.Explainer(xgb)
```

Code Cell 46

```
shap_values = explainer(X_test)
```

Code Cell 47

```
shap.plots.beeswarm(shap_values)
```

Code Cell 48

```
shap.plots.bar(shap_values)
```

Markdown Cell 49

SHAP Insights

- Customers with high monthly charges are more likely to churn.
- Customers with short tenure have higher churn probability.
- Long-term contracts reduce customer churn.
- Internet service type strongly affects customer retention.

Code Cell 50

```
import pickle
```

Code Cell 51

```
pickle.dump(xgb, open("../models/churn_model.pkl", "wb"))
```

Code Cell 52
